



LECTURE

Preparing Data for RAG Solutions



Why is Data Prep Important for RAG?

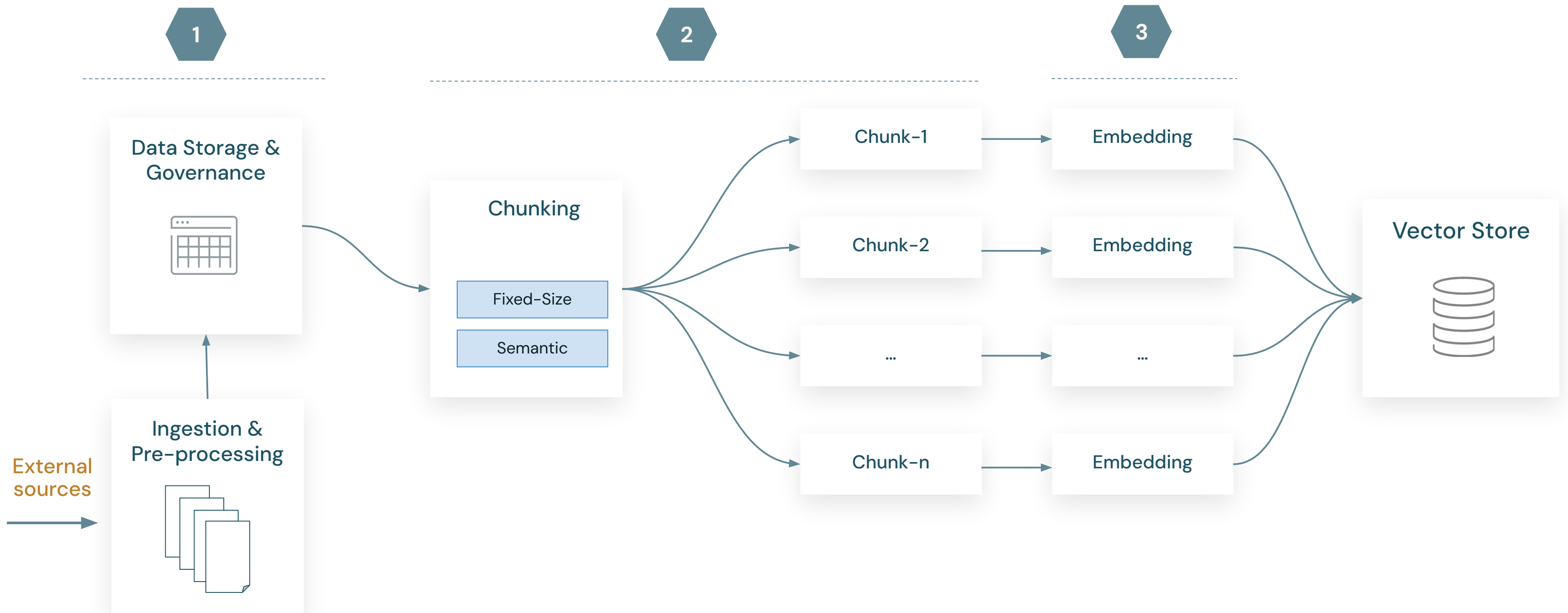
Potential issues when data is prepared improperly

- **Poor quality model output:** If data is inaccurate, incomplete, or biased, the RAG system is more likely to produce misleading or incorrect responses.
- **“Lost in the middle”:** In long context, LLMs tend to overlook the documents placed in the middle ([Related Research Paper](#) and [needle in haystack test](#) repo).
- **Inefficient retrieval:** Poorly prepared data would decrease the accuracy and precision of retrieving relevant information from knowledge base.
- **Exposing data:** Poor data governance could lead to exposing data during the retrieval process.
- **Wrong embedding model:** Wrong embedding model would decrease the quality of embeddings and retrieval accuracy.



Data Prep Process Overview

A simple data prep process

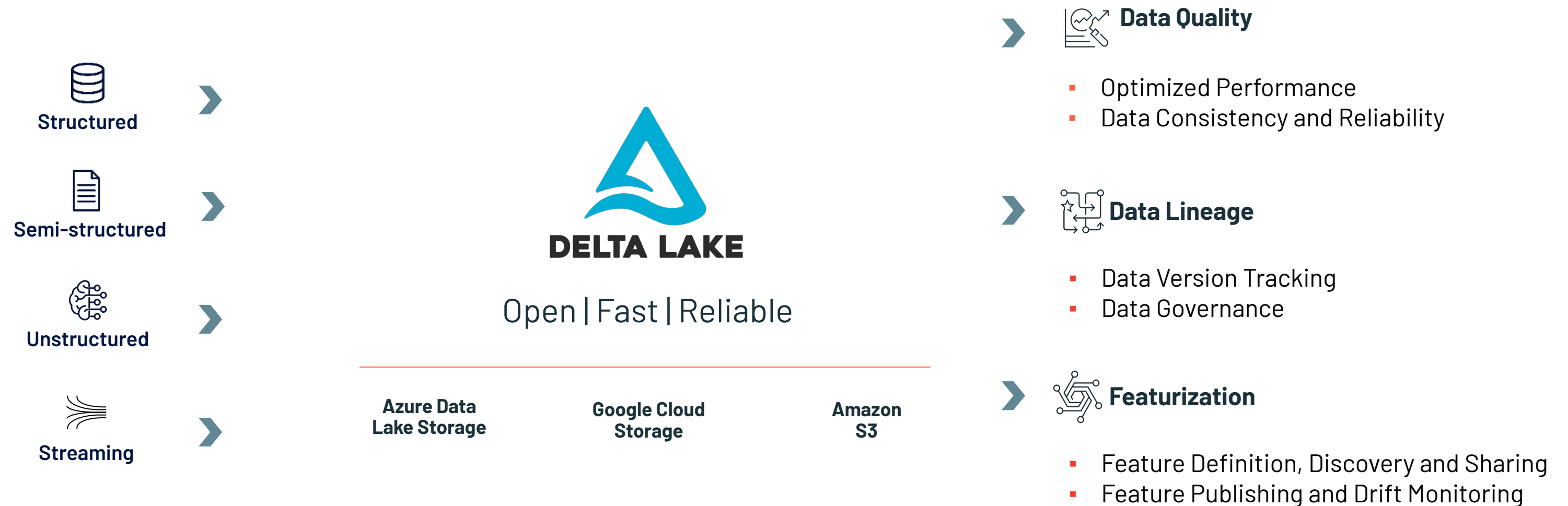


1.Data Storage and Governance



What is Delta Lake?

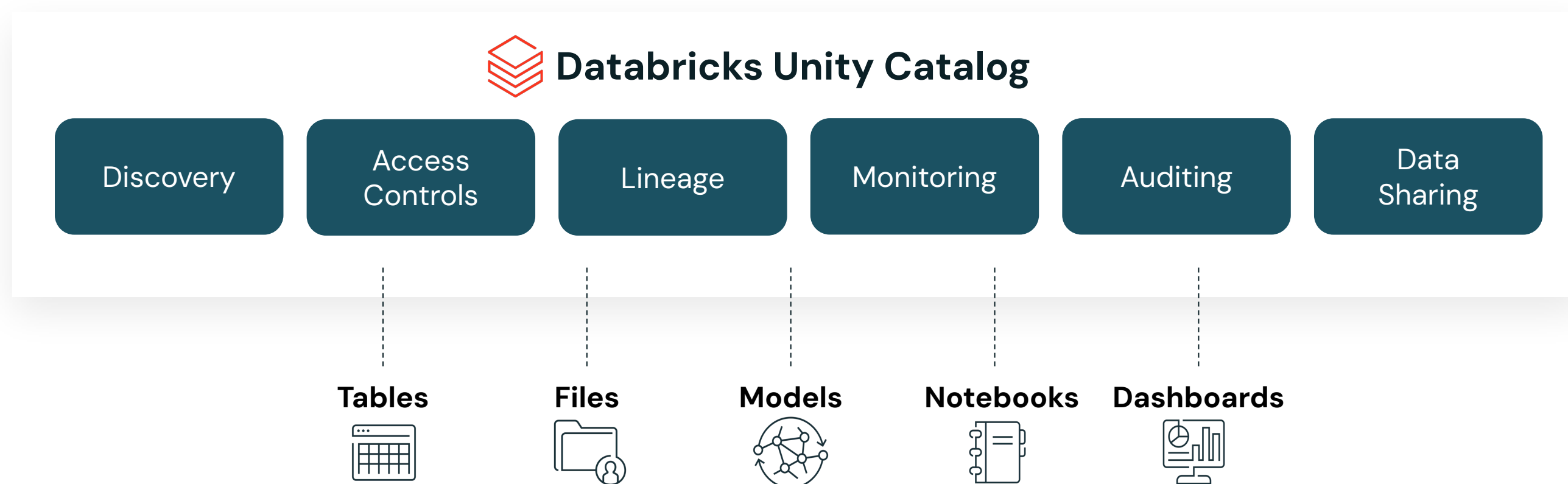
- A **unified data management layer** that brings data reliability and fast analytics to cloud data lakes. It is the optimized storage layer that provides the foundation for storing data and tables in the Databricks DI Platform.



Unity Catalog (UC)

Governance, discovery, access control for RAG applications

- Unified visibility into data and AI
- Single permission model for data and AI
- Open data sharing

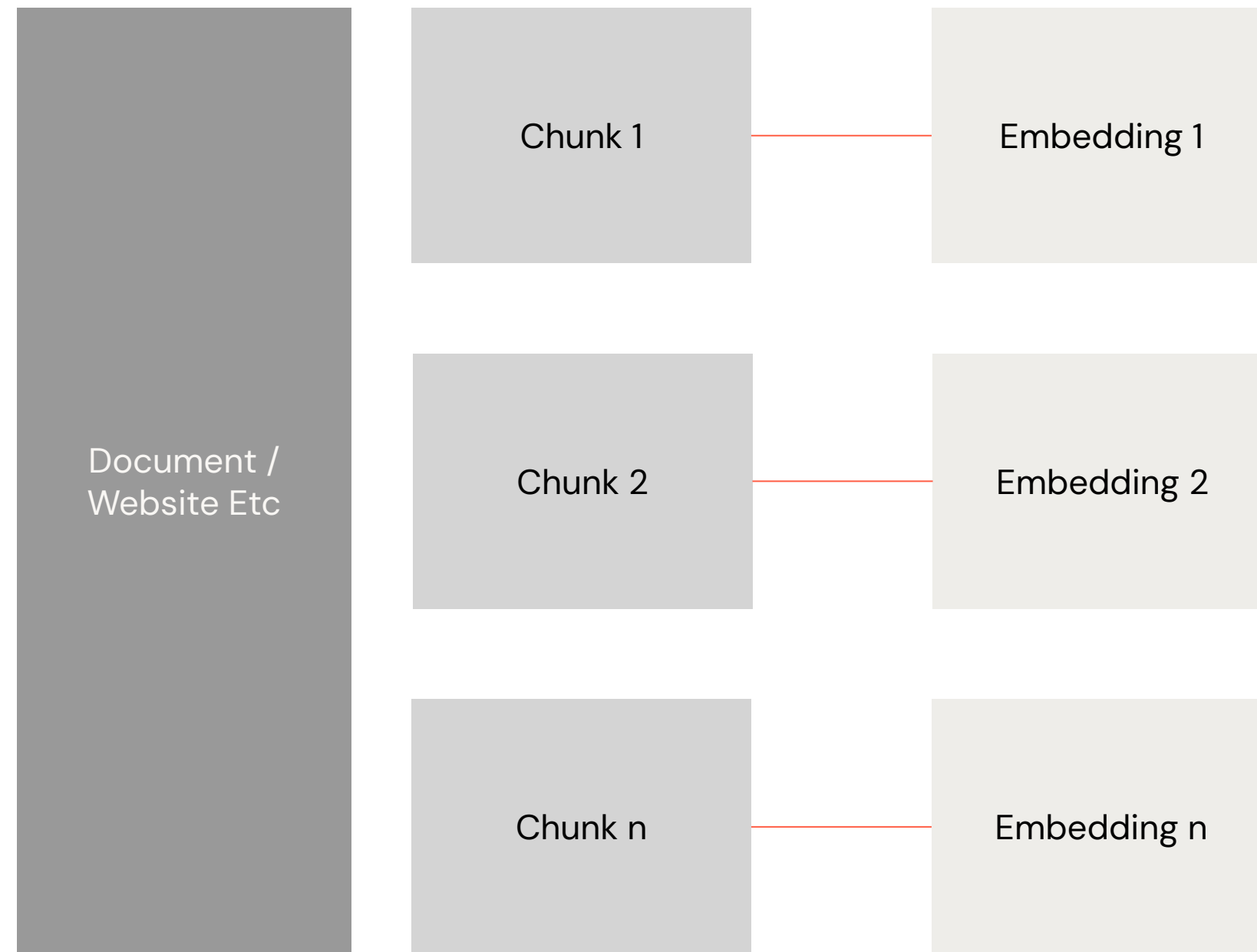


02. Data Extraction & Chunking



Data/Document Extraction

Ingesting (text) documents and making them searchable



Typical Process:

1. Split documents into chunks
2. Embed the chunks with a model
3. Store them in a vector store

Constraints/Risks:

- Chunks could be out of Context
- No overall logic/rigor



How to Chunk Data?

How should we organise it?

Neural network

[Article](#)[Talk](#)

Read

Edit

View history

Tools

From Wikipedia, the free encyclopedia

For other uses, see *Neural network* (disambiguation).

A **neural network** can refer to *either* a neural circuit of biological neurons (sometimes also called a *biological neural network*), or a network of artificial neurons or nodes in the case of an *artificial neural network*.^[1] Artificial neural networks are used for solving artificial intelligence (AI) problems; they model connections of biological neurons as weights between nodes. A positive weight reflects an excitatory connection, while negative values mean inhibitory connections. All inputs are modified by a weight and summed. This activity is referred to as a linear combination. Finally, an activation function controls the amplitude of the output. For example, an acceptable range of output is usually between 0 and 1, or it could be −1 and 1.

These artificial networks may be used for predictive modeling, adaptive control and applications where they can be trained via a dataset. Self-learning resulting from experience can occur within networks, which can derive conclusions from a complex and seemingly unrelated set of information.^[2]

Overview

[edit]

A biological neural network is composed of a group of chemically connected or functionally associated neurons. A single neuron may be connected to many other neurons and the total number of neurons and connections in a network may be extensive. Connections, called synapses, are usually formed from axons to dendrites, though dendrodendritic synapses^[3] and other connections are possible. Apart from electrical signalling, there are other forms of signalling that arise from neurotransmitter diffusion.

Artificial intelligence, cognitive modelling, and neural networks are information processing paradigms inspired by how biological neural systems process data. Artificial intelligence and cognitive modelling try to simulate some properties of biological neural networks. In the artificial intelligence field, artificial neural networks have been applied successfully to speech recognition, image analysis and adaptive control, in order to construct software agents (in computer and video games) or autonomous robots.

Historically, digital computers evolved from the von Neumann model, and operate via the execution of explicit instructions via access to memory by a number of processors. On the other hand, the origins of neural networks are based on efforts to model information processing in biological systems. Unlike the von Neumann model, neural network computing does not separate memory and processing.

Neural network theory has served to identify better how the neurons in the brain function and provide the basis for efforts to create artificial intelligence.

History

[edit]

The preliminary theoretical base for contemporary neural networks was independently proposed by Alexander Bain^[4] (1873) and William James^[5] (1890). In their work, both thoughts and body activity resulted from interactions among neurons within the brain.

For Bain,^[4] every activity led to the firing of a certain set of neurons. When activities were repeated, the connections between those neurons strengthened. According to his theory, this repetition was what led to the formation of memory. The general scientific community at the time was skeptical of Bain's^[4] theory because it required what appeared to be an inordinate number of neural connections within the brain. It is now apparent that the brain is exceedingly complex and that the same brain "wiring" can handle multiple problems and inputs.

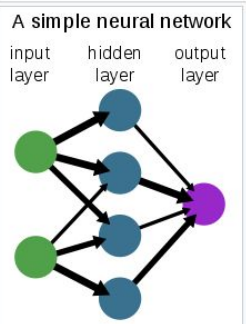
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A simple neural network

input layer

hidden layer

output layer



Simplified view of a feedforward artificial neural network

Title

Section

Diagram

Semantic Chunking:

- Chunk by sentence/paragraph/section
- Leverage special punctuation (i.e. '.', '\n')
- Include/Inject metadata/tags/title(s)

&/OR

Fixed-size Chunking:

- Divide by a specific number of tokens
- Simple and computationally cheap method

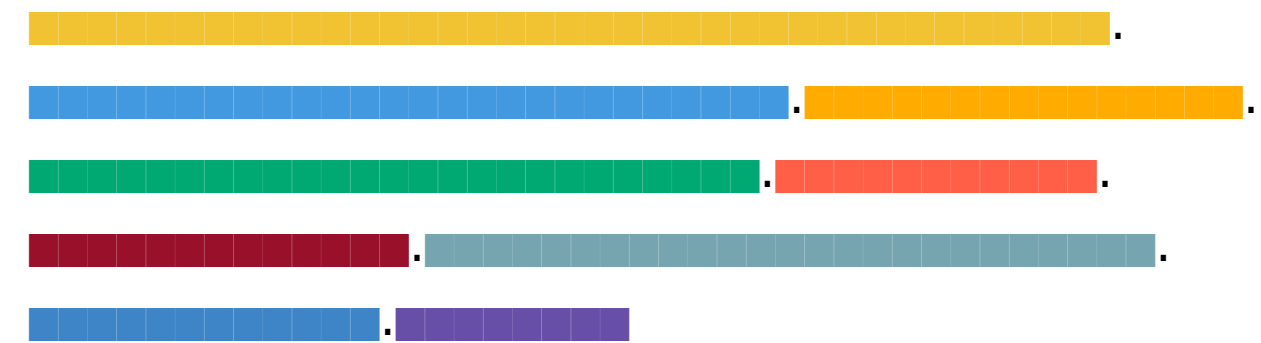
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Chunking Strategy is Use-Case Specific

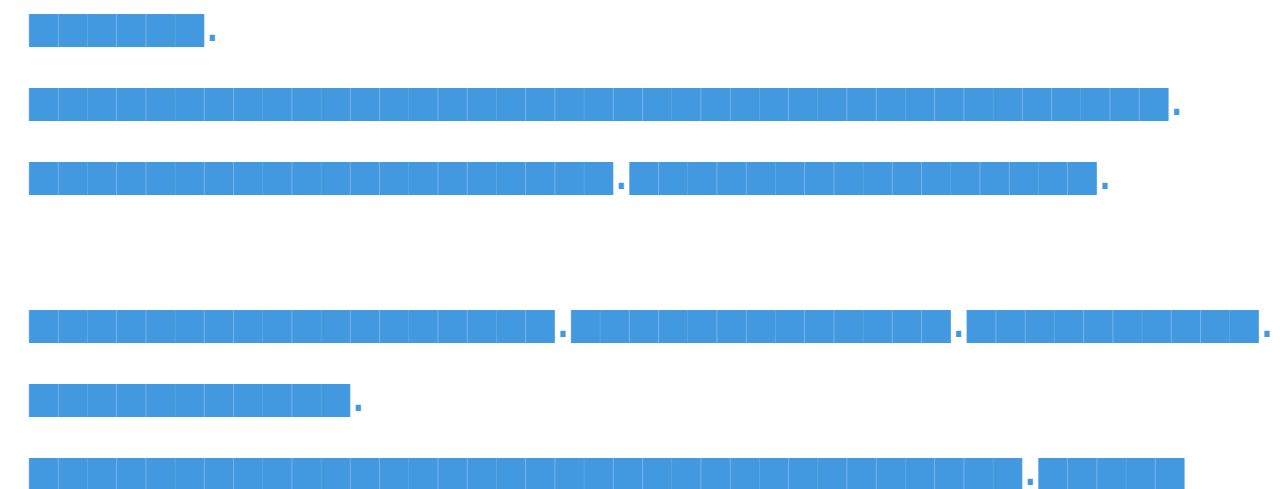
Another iterative step! Experiment with different chunk sizes and approaches

- How long are our documents?
 - 1 sentence?
 - N sentences?
- If 1 chunk = **1 sentence**, embeddings focus on specific meaning
- If 1 chunk = **multiple paragraphs**, embeddings capture broader theme
 - How about splitting by headers?

Chunking by sentence:



Chunking by Paragraph:

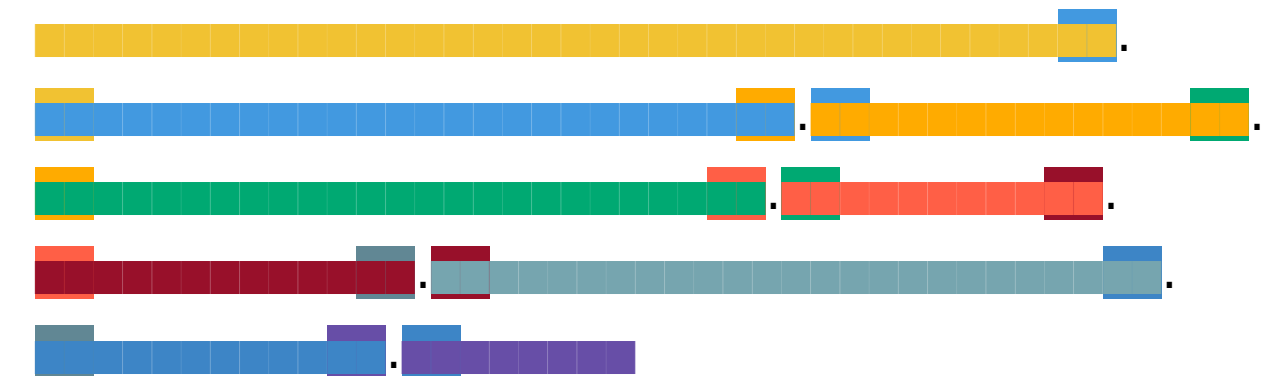


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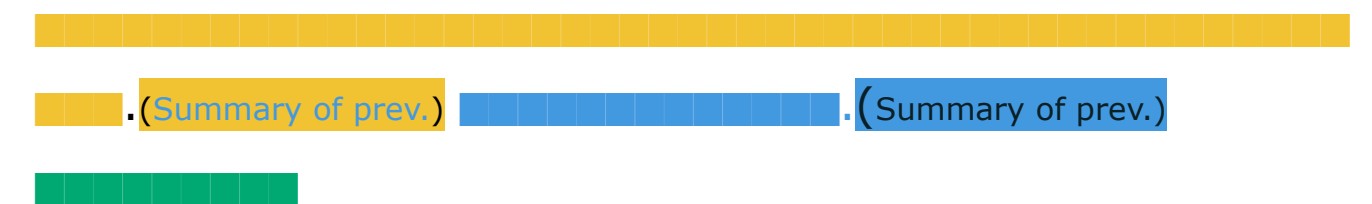
Another iterative step! Experiment with different chunk sizes and approaches

- Chunk **overlap** defines the amount of overlap between consecutive chunks, ensuring that no contextual information is lost between them.
- **Windowed summarization** is a 'context-enriching' chunking method where each chunk includes a 'windowed summary' of previous few chunks.

Chunk overlap:



Windowed summarization:



- Prior knowledge of user's query patterns can be helpful (*i.e. query length?*)
 - While long queries may have better aligned embeddings to returned chunks, shorter queries could be more precise



Advanced Chunking Strategies

Summarization

Neural network

Article Talk

From Wikipedia, the free encyclopedia

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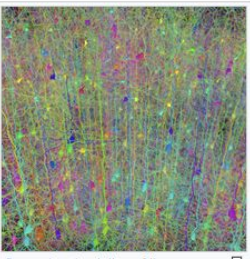
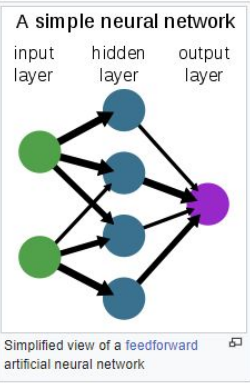
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Section A

Summarize with LLM

Section B

Summarize with LLM

Section C

Summarize with LLM



Advanced Chunking Strategies

Summarization with metadata

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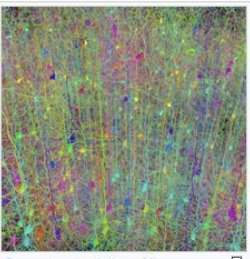
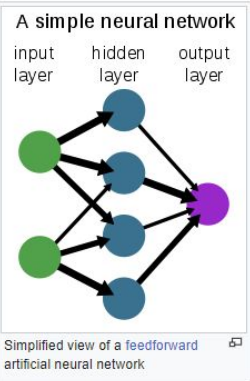
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Section A

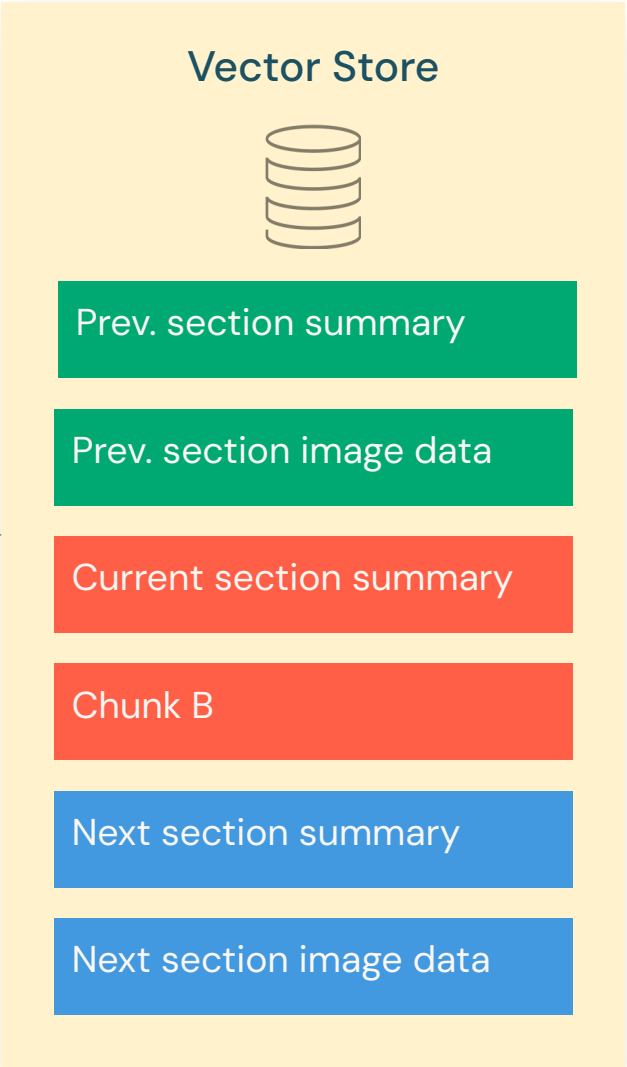
Summarize with LLM

Section B

Summarize with LLM

Section C

Summarize with LLM



Data Extraction and Chunking Challenges

Working with complex documents

HOLIDAY PACKAGES

Our exclusive range of holiday packages have been specially designed with you in mind and feature a choice of accommodation and local experiences. These packages are a perfect introduction to Bali and offer great value for money.

Relax on a beautiful beach, explore lush green rice terraces, or visit the local markets. Whether you're looking for adventure and fun for the whole family or a quiet, romantic hideaway, Bali is the perfect holiday destination.

Be inspired by one of our fantastic holiday packages and discover all that Bali has to offer.



UBUD SPA & WELLNESS RETREAT

4 NIGHTS

The Royal Pita Maha is a haven for personal wellness where you can indulge at the day spa or join a yoga or meditation session, perfect for those who wish to recharge and feel refreshed and rejuvenated.

Spend a day at The Yoga Dham, located in the heart of Ubud, and experience the holistic healing at this full service yoga studio.

INCLUDES

- 4 nights 5 star accommodation in a Deluxe Pool Villa at The Royal Pita Maha Resort
- Full breakfast daily
- Indonesian set menu dinner on one evening (includes beverages)
- Afternoon tea at Royal Kadisea Restaurant daily
- Yoga class and scheduled cultural activities daily
- Yoga and Balinese I Chai Experience
- Shuttle services to Ubud centre
- Welcome drink, fruit basket and gift on arrival
- Return private car transfers from Ngurah Rai International Airport

Refer to page 73 for more details on this property.

From \$1055⁺⁺ per person both ways (USD)

⁺⁺Price includes 1 free night and 10% early bird discount (book 45 days prior to travel), based on 4 night package, valid 1 Apr - 30 Jun, 1 Sep - 19 Dec 20, 11 Jan - 31 Mar 21. Ask your travel agent for prices for other dates and room types.



BEST OF BALI BEACHES

7 NIGHTS

Enjoy the sun and sand at Legian and Candidasa beaches for the ultimate Bali beach getaway.

With fantastic surf right on your doorstep and a relaxed vibe, Legian Beach Hotel has a wide range of facilities and a great beachfront location. Spend an afternoon on a relaxing sunset dinner cruise with live entertainment and a buffet dinner.

A two hour drive from Legian, you'll arrive at Candidasa, the perfect base to explore western Bali's villages and countryside. Clear water and coral reef makes this destination highly popular with divers.

INCLUDES

- 4 nights 4 star accommodation in a Deluxe room at Legian Beach Hotel
- 3 nights 4 star accommodation in a Deluxe Garden room at Cand Beach Resort
- 6 Eps • Full breakfast daily
- 3 hour Sunset Dinner Cruise from Legian
- Daily guided morning walks from Cand Beach Resort & Spa
- Return private car transfers from Ngurah Rai International Airport
- Private car transfers between Legian Beach Hotel and Cand Beach Resort & Spa

Refer to pages 71 and 74 for more details on these properties.

From \$935⁺⁺ per person both ways (USD, 202)

⁺⁺Based on 7 night package, valid 1 Apr - 14 Jun, 15 Oct - 22 Dec 20, 6 Jan - 21 Mar 21. Ask your travel agent for prices for other dates and room types.



Image

Text

Table

Price and disclaimer

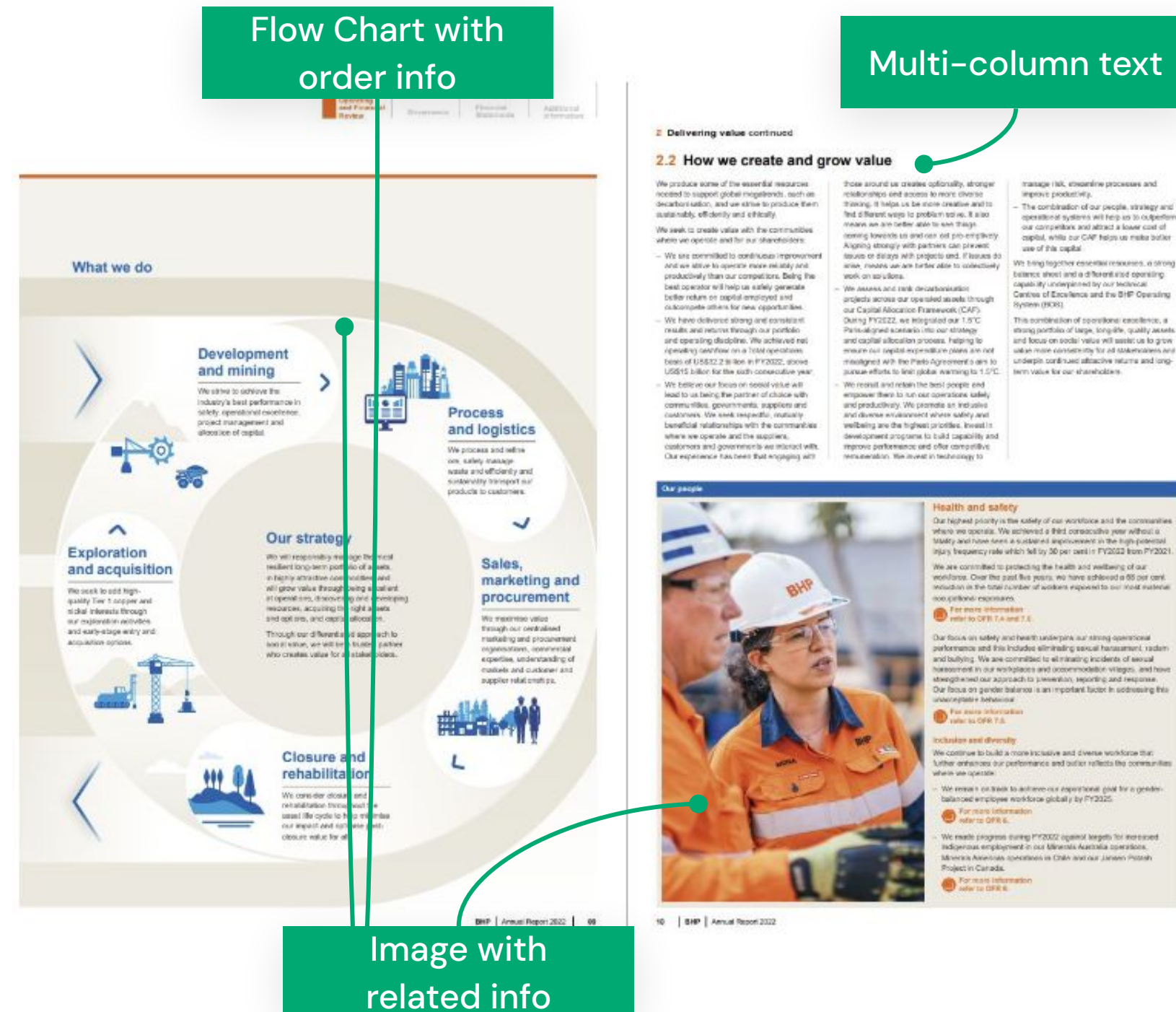
Other challenges:

- Text mixed with image
- Irregular placement of text
- Color encoded focus (*Important for context*)



Data Extraction and Chunking Challenges

Working with complex documents



Other challenges:

- Chart with hierarchical information. Keeping the order of the information is critical.
- Multi-column text and the order of columns if crucial.
- Keeping images with related information is crucial.



Data Extraction and Chunking Challenges

Working with different document types

- Preserving logical sections of the content might be difficult for some document types.
 - HTML requires tag based chunking to preserve logical structure of the content.
- HTML Tables are token heavy.
- Example: 3x3 small table;
 - Plain text: 11 tokens
 - JSONL: 29 tokens
 - HTML: 132 tokens (!!)

```
html
<table style="border-collapse: collapse; width: 50%;">
  <tr>
    <th style="background-color: #f2f2f2; padding: 8px;">Header 1</th>
    <th style="background-color: #f2f2f2; padding: 8px;">Header 2</th>
    <th style="background-color: #f2f2f2; padding: 8px;">Header 3</th>
  </tr>
  <tr>
    <td style="border: 1px solid #ddd; padding: 8px;">Row 1, Cell 1</td>
    <td style="border: 1px solid #ddd; padding: 8px;">Row 1, Cell 2</td>
    <td style="border: 1px solid #ddd; padding: 8px;">Row 1, Cell 3</td>
  </tr>
  <tr>
    <td style="border: 1px solid #ddd; padding: 8px;">Row 2, Cell 1</td>
    <td style="border: 1px solid #ddd; padding: 8px;">Row 2, Cell 2</td>
    <td style="border: 1px solid #ddd; padding: 8px;">Row 2, Cell 3</td>
  </tr>
  <tr>
    <td style="border: 1px solid #ddd; padding: 8px;">Row 3, Cell 1</td>
    <td style="border: 1px solid #ddd; padding: 8px;">Row 3, Cell 2</td>
    <td style="border: 1px solid #ddd; padding: 8px;">Row 3, Cell 3</td>
  </tr>
</table>
```

```
json
{
  "table": [
    {
      "Header 1": "Row 1, Cell 1",
      "Header 2": "Row 1, Cell 2",
      "Header 3": "Row 1, Cell 3"
    },
    {
      "Header 1": "Row 2, Cell 1",
      "Header 2": "Row 2, Cell 2",
      "Header 3": "Row 2, Cell 3"
    },
    {
      "Header 1": "Row 3, Cell 1",
      "Header 2": "Row 3, Cell 2",
      "Header 3": "Row 3, Cell 3"
    }
  ]
}
```

```
text
Header 1    Header 2    Header 3
Row 1, Cell 1  Row 1, Cell 2  Row 1, Cell 3
Row 2, Cell 1  Row 2, Cell 2  Row 2, Cell 3
Row 3, Cell 1  Row 3, Cell 2  Row 3, Cell 3
```



General Approaches

Approaches to address unstructured/complex raw text documents

Traditional Approach

Libraries:

- PyMuPDF

Features:

- Breaks down text to into raw constructs
- Very low level requires hard coding rules

Use a layout model^(*)

Libraries:

- Hugging Face's LayoutLMv3
- doctr
- PyPDF
- Unstructured

Features:

- Apply Deep learning models built to do text extraction and context extraction

Multi-Modal Models

Models:

- OpenAI's GPT-4o (and beyond)
- Alphabet's Gemini1.5 (and beyond)
- Other OSS models (i.e. Dolphin's Series, OpenFlamingo, Llava, OLMo)

Features:

- Multimodal LLMs intrinsically understand images but are still more experimental at this stage

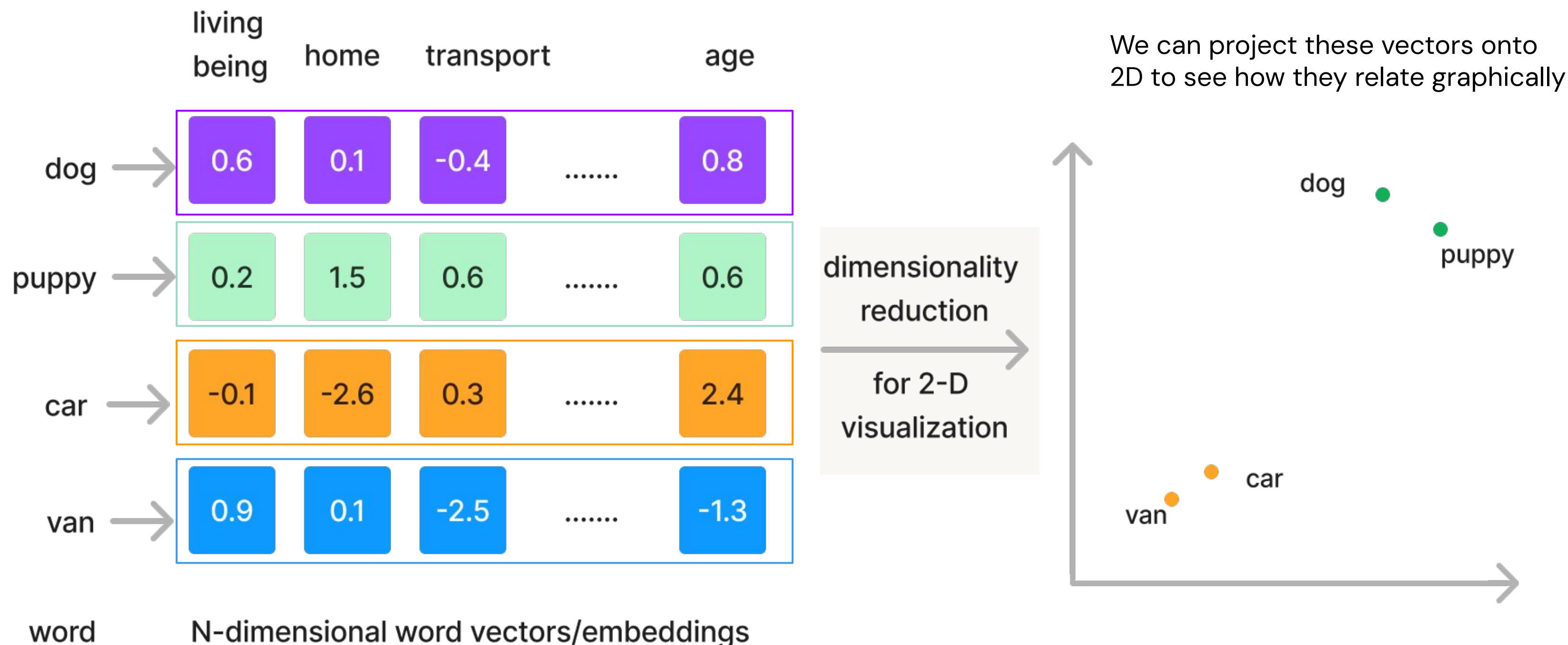


03. Embedding Model



Refresher: Representing Words with Vectors

Embedding: A numerical representation of content



Embedding Models

Choosing the **right model for your application**

- Data/Text properties:
 - Vocabulary size in your text/documents (some models handles more diverse words)
 - Domain/Topic (i.e. finance, medical, news etc.)
 - Text length: typical length of chunks/docs to be embedded
- Model capabilities:
 - Multi-Language support
 - Embedding dimensions/size: more storage cost for higher dimensions

Practical considerations:

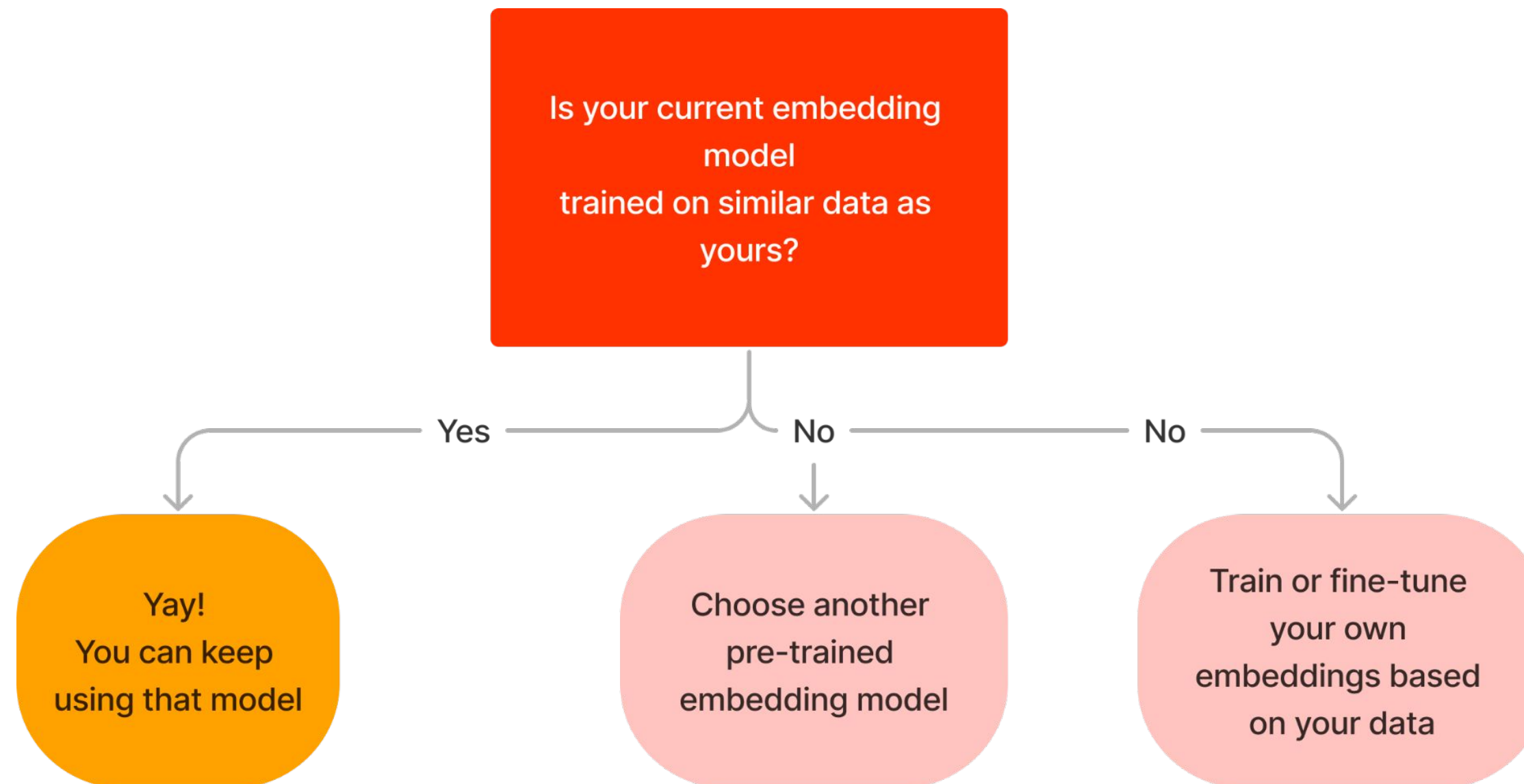
- Be aware of context window limitations. Many embedding models will **ignore text beyond their context window limits**.
- Privacy and cost/licensing when using proprietary API-based models.

⇒ benchmark multiple models & choose the one that strikes the best balance.



Tip 1: Choose Your Embedding Model Wisely

The embedding model should represent BOTH queries and documents

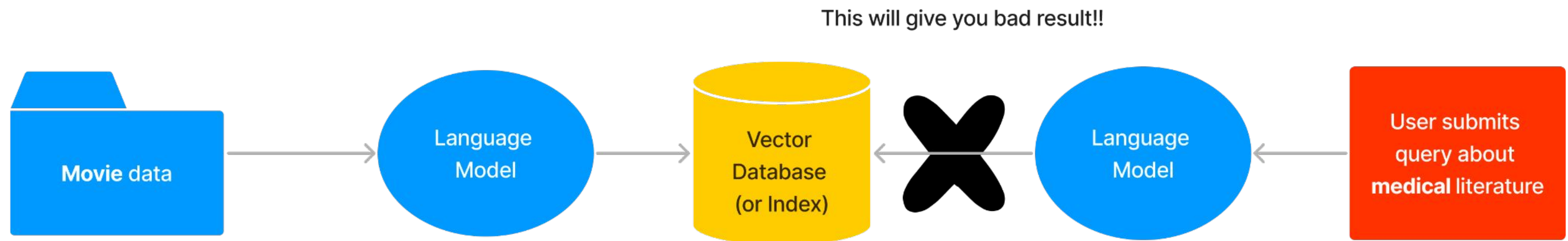


This practice has been around for years in NLP.
Example: Fine-tune BERT embeddings



Tip 2: Ensure similar Embedding Space for both Queries and Documents

- Use the same embedding model for indexing and querying
- OR – if you use different embedding models, make sure they are trained on similar data (therefore produce the same embedding space!)

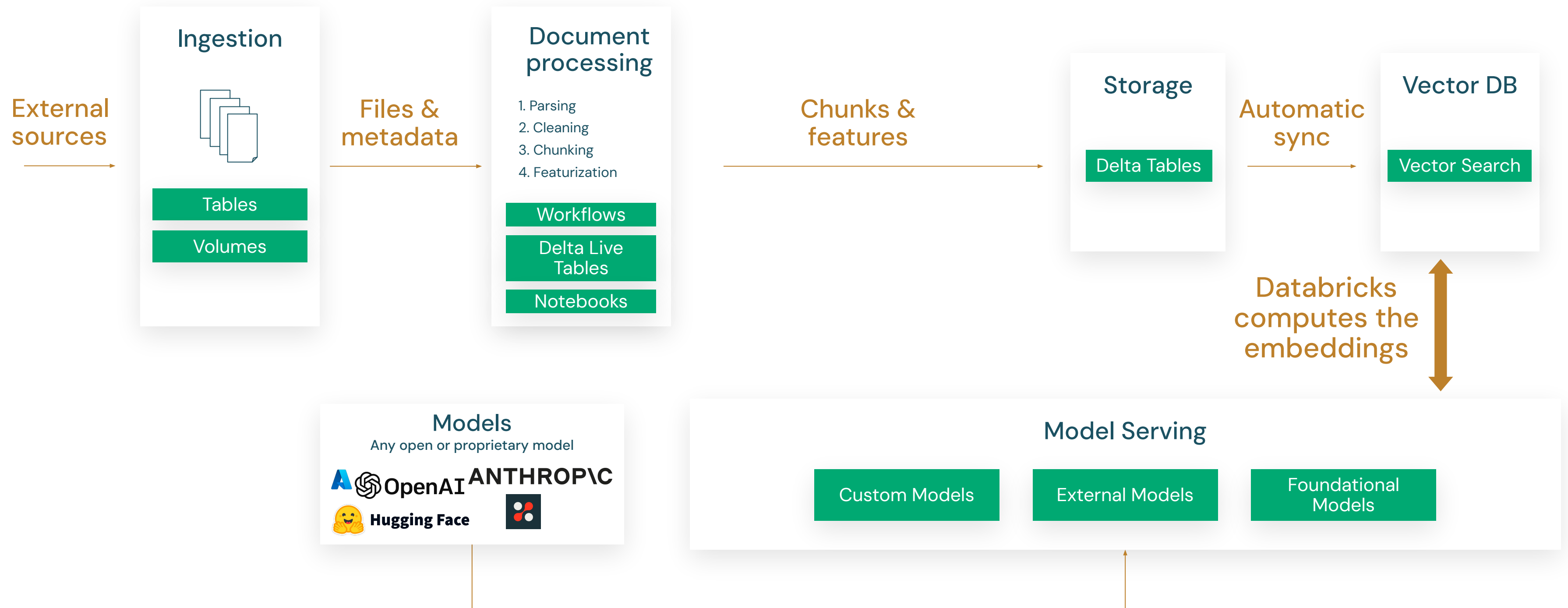


Data Preparation in Databricks



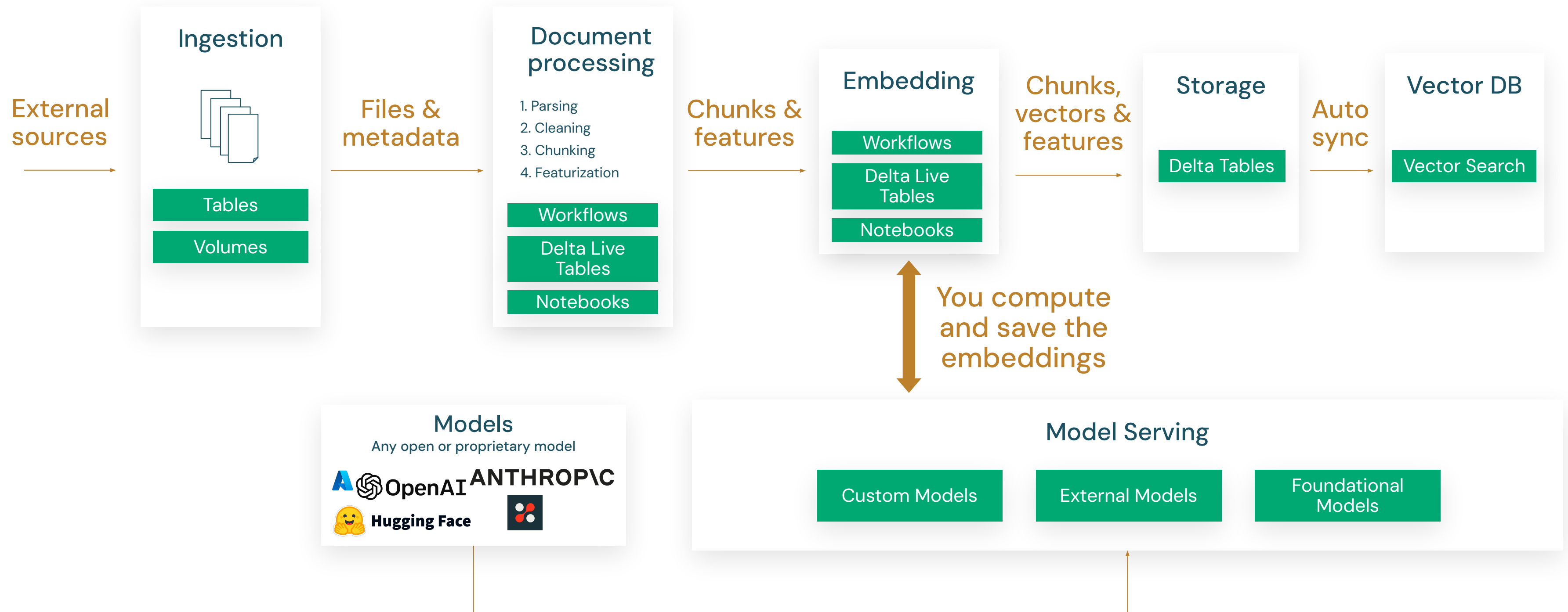
Unstructured Data Prep

Vector Search with Databricks-managed embeddings



Unstructured Data Prep

Vector Search with user-managed embeddings



Structured Data Prep

Feature Serving and Online Tables

